

and stable operation under heavy loads and high-temperature conditions. Engineered by Magnachip for data centres, industrial systems, and solar inverters, the MOSFETs feature fast anti-parallel diode functionality to reduce residual current during switching transitions.

Magnachip Semiconductor
<https://www.magnachip.com>

Hydrogen sensors

Hydrogen sensors are emerging as a key safety technology for electric vehicles by enabling early detection of battery failures before thermal runaway occurs. Recent developments from companies including Honeywell and Posifa

Technologies focus on detecting hydrogen released during the initial venting stage of lithium-ion batteries, allowing faster preventive action. Research conducted in Hangzhou, China, has also highlighted room-temperature resistive hydrogen sensors for integration inside battery packs. These compact sensors improve EV safety, support battery management systems, and are increasingly targeted at energy storage, automotive, and backup power applications.

Posifa Technologies
<https://posifatech.com>

High-power violet laser diode

Nuvoton Technology has introduced a high-power violet laser diode for maskless lithography in advanced semiconductor packaging applications.



Operating at 402nm wavelength with high optical output, the device is designed to improve throughput, expand material compatibility, and support next-generation semiconductor manufacturing driven by AI and high-performance computing demand. The laser diode enables faster direct imaging in maskless lithography systems, helping reduce development cycles and improve handling of substrate

distortion and warpage. Packaged in a compact TO-9 CAN form factor, the device supports integration into industrial photolithography and optical processing systems.

Nuvoton Technology
<https://www.nuvoton.com.cn>

Non-contact displacement sensors

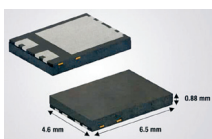
Kaman Precision Products, Inc. has introduced a new family of non-contact displacement sensors designed for high-precision condition monitoring in harsh industrial environments. The sensors deliver micron-level measurement resolution and operate in high-pressure conditions, enabling accurate monitoring of vibration, shaft runout, and rotational speed in critical machinery. Built with a non-contact sensing approach, the devices reduce mechanical wear while maintaining long-term measurement accuracy. The IP67-rated sensors support multiple output formats for integration with industrial control systems and can be customised for OEM applications. The technology also supports gear monitoring, thickness inspection, and defect detection in demanding industrial operations.



Kaman Precision Products Measurement Division
www.kamansensors.com

Compact ultrafast rectifiers

Vishay Intertechnology has introduced 16 new 200V FRED Pt ultrafast rectifiers in a compact DFN6546A package



for automotive, industrial, and consumer power applications. The devices are designed to improve power conversion efficiency while reducing board space in high-frequency circuits such as DC/DC converters, inverters, and protection systems. The rectifiers combine low forward voltage, fast switching performance, and reduced power losses with improved thermal

capability in a low-profile package. Automotive-qualified versions support applications including EV power platforms, ADAS, lighting, battery management, and onboard chargers. The devices also feature wettable flanks for simplified inspection and compatibility with automated manufacturing processes.

Vishay Intertechnology
<https://www.vishay.com>

Microcontrollers with integrated logic blocks

Microchip Technology has introduced a new class of microcontrollers that integrate configurable logic blocks directly into the microcontroller architecture, enabling hardware-accelerated control



for timing-critical embedded applications. The devices target motor control, industrial automation, and automotive systems where deterministic performance and low latency are essential. By combining programmable logic with embedded control on a single chip, the microcontrollers reduce the need for external logic components, lowering board space, system complexity, and power consumption. The architecture supports parallel signal processing, predictable startup behaviour, and hardware-based timing for improved responsiveness. Integrated security features, timing analysis tools, and a graphical drag-and-drop configuration environment further simplify development and accelerate deployment of real-time embedded systems.

Microchip Technology
www.microchip.com

Dual 3D V-Cache processor

AMD has introduced the Ryzen 9 9950X3D2 Dual Edition processor,



bringing dual 3D V-Cache technology to a desktop CPU for high-performance gaming, content creation, and development workloads. Built on the Zen 5 architecture, the processor combines expanded